Lecture 12: Managing Portfolio Risk

Questions

Work through each problem before reading the solutions. Keep full precision until your final answer.

Use nominal annual rates and semiannual payments. Apply the benchmark observed at the start of each semester. Debt and swaps have matching notionals, currencies, benchmarks and dates. Ignore fees, taxes and default. Net swap cash flow means received minus paid.

1. Two firms and one bank

Firm A has 7% fixed debt, pays SOFR + 1% to a bank and receives 5% fixed. Firm B has debt at SOFR + 2%, pays 6% fixed to the same bank and receives SOFR. Both swaps have equal notionals and matching dates. What are A's final annual borrowing rate, B's final annual borrowing rate and the bank's annual net spread on one common notional?

A. A: SOFR + 3%; B: 8% fixed; bank: 2% fixed

B. A: SOFR + 2%; B: 8% fixed; bank: 1% fixed

C. A: 3% fixed; B: SOFR + 8%; bank: 2% fixed

D. A: SOFR + 3%; B: 4% fixed; bank: SOFR + 2%

2. Fix a Mexican bond's income

A portfolio holds an MXN 2,000 bond bought at par. It pays TIIE + 1% annually, semiannually, and matures in two years. A matching IRS receives 7% fixed and pays TIIE. What are the combined annual interest income rate and the combined interest receipt each semester, excluding principal?

A. 7% fixed; MXN 70

B. 8% fixed; MXN 160

C. 8% fixed; MXN 80

D. TIIE + 8%; a variable receipt

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3. Rates fall: select the correct reset

Use the MXN 2,000 bond and receive-7%-fixed/pay-TIIE swap from Question 2. TIIE is 7% at t=0, 4% at t=0.5 and 3% at t=1.0. Each payment uses the preceding observation. At t=1.0, what are the bond coupon, the net IRS cash flow and the combined receipt?

A. MXN 40; +MXN 40; MXN 80

B. MXN 50; +MXN 30; MXN 80

C. MXN 50; -MXN 30; MXN 20

D. MXN 100; +MXN 60; MXN 160

4. Rates rise: compare the entire portfolio

Keep the bond and swap from Question 2. TIIE observed at t=0, 0.5, 1.0, 1.5 and 2.0 is 7%, 9%, 11%, 11% and 12%, respectively. The final observation is unused. Over two years, what are total bond interest without the IRS, total interest with the IRS, and the combined final cash flow at t=2.0 including principal? Assume no reinvestment.

A. MXN 420; MXN 420; MXN 2,120

B. MXN 320; MXN 420; MXN 2,080

C. MXN 420; MXN 320; MXN 80

D. MXN 420; MXN 320; MXN 2,080

5. Positive pesos, negative dollars

The hedged portfolio in Question 2 invests MXN 2,000 at t=0 and accumulates MXN 2,320 at t=2, including principal. Hold all coupons in MXN without reinvestment and convert everything at t=2. USDMXN, measured in MXN per USD, rises from 20 to 25. What is the cumulative USD return and the terminal USDMXN above which that return becomes negative?

A. +16.0%; 25.0

B. -7.2%; 23.2

C. -20.0%; 20.0

D. +45.0%; 23.2

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Worked solutions

1. Two firms and one bank - Answer A

Step 1. Combine A's debt and swap. I add what A pays and subtract what it receives: 7% + (SOFR + 1%) - 5% = SOFR + 3%.

Step 2. Do the same for B. Its final rate is (SOFR + 2%) + 6% - SOFR = 8% fixed.

Step 3. Reverse the client legs for the bank. The bank receives SOFR + 1% and 6%, and pays 5% and SOFR. Its net annual spread is 2%. SOFR cancels because the notionals, dates and benchmarks match.

Interpretation. The firms change their exposure; the bank retains a fixed spread before costs and default losses. No direct alternative borrowing quotes were supplied, so these results do not establish savings.

Check your reasoning. The bank's spread is a rate on one common notional, not a dollar amount and not a rate on the sum of both notionals.

2. Fix a Mexican bond's income - Answer C

Step 1. Identify the existing receipt. The bond pays TIIE + 1%. The manager wants to remove the uncertainty in this floating income.

Step 2. Add the IRS cash flows. The manager receives 7% and pays TIIE. Combined annual income is (TIIE + 1%) + 7% - TIIE = 8% fixed.

Step 3. Convert the annual rate to a semester. MXN 2,000 x 8% / 2 = MXN 80. Add the bond's principal only on the maturity date.

Interpretation. The swap fixes scheduled peso income. It does not fix the investment's USD return.

Check your reasoning. An investor adds net swap receipts to bond income. Do not apply the borrower's cost formula to investment receipts.

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3. Rates fall: select the correct reset - Answer B

Step 1. Select the rate for this semester. The t=1.0 payment uses 4% observed at t=0.5. The new 3% observation sets the next payment.

Step 2. Compute the bond coupon. MXN 2,000 x (4% + 1%) / 2 = MXN 50.

Step 3. Compute the swap and combine. Receive MXN 2,000 x 7% / 2 = 70; pay MXN 2,000 x 4% / 2 = 40. Net IRS = 70 - 40 = +30. Combined receipt = 50 + 30 = 80.

Interpretation. The lower coupon is offset by a positive swap receipt. The manager still receives the locked MXN 80.

Check your reasoning. The rate observed on the payment date does not determine the payment for the semester that just ended.

4. Rates rise: compare the entire portfolio - Answer D

Step 1. Calculate four bond coupons. Using the start-of-period resets gives 80, 100, 120 and 120 pesos. Total unhedged interest is MXN 420.

Step 2. Calculate the swap nets. Fixed receipts are MXN 70 each. Floating payments are 70, 90, 110 and 110. Net swap cash flows are 0, -20, -40 and -40; their sum is -100.

Step 3. Add the cash flows. Combined interest is 420 - 100 = MXN 320, or 80 each semester. At maturity, the bond pays 2,000 + 120 and the swap pays a net 40. Combined final cash flow is 2,080.

Interpretation. The manager gives up MXN 100 of upside in return for predictable income. A negative swap cash flow is consistent with a successful hedge.

Check your reasoning. Keep interest income separate from principal repayment. The swap does not repay a second principal amount.

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5. Positive pesos, negative dollars - Answer B

Step 1. Convert the initial investment. MXN 2,000 / 20 = USD 100.

Step 2. Convert the accumulated proceeds. MXN 2,320 / 25 = USD 92.80. Cumulative USD return = 92.80 / 100 - 1 = -7.2%.

Step 3. Find the break-even exchange rate. The local gross return is 2,320 / 2,000 = 1.16. USD return = 1.16 x 20 / USDMXN at t=2 - 1. Set it to zero: terminal USDMXN = 23.2. Above 23.2, the dollar return is negative.

Interpretation. Peso depreciation can reverse a positive local return. The IRS fixes peso interest income, while currency exposure remains.

Check your reasoning. USDMXN is pesos per dollar, so divide pesos by the quote. The single terminal-rate threshold assumes that all coupons are converted only at the end.